Expressions and equivalent forms

Explore and build the idea

20 daily questions include C1 and C2. Use separate paper when helpful.

Lesson focus: Write expressions and show why two forms have the same value.

R1. 3 markers cost $1.95. Find dollars per marker.

R2. Find $4 \div \frac{2}{3}$.

R3. Find $288 \div 24$.

R4. 18 shells are 30% of a collection. Find the whole collection.

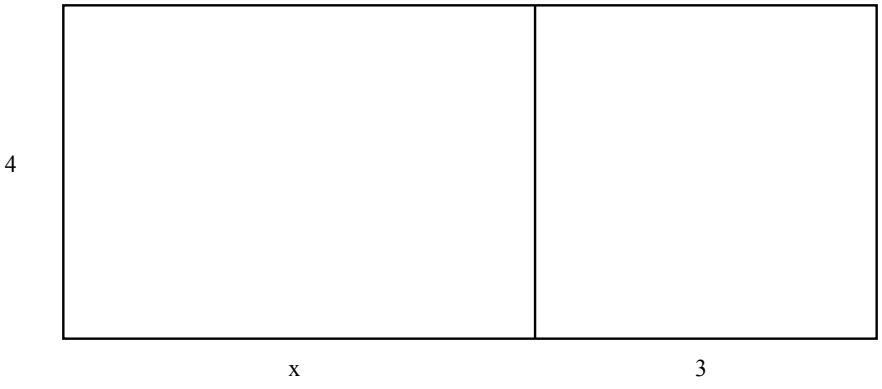
P1. A sticker sheet costs n dollars. Write the cost of 5 sheets. Find it when n = 2 and when n = 4.

P2. Find $2a + 5$ when $a = 6$. Write the multiplication before adding.

P3. Find 4^2, 4×2, 3^3, and $(\frac{2}{3})^2$. Which pair has different meanings?

P4. A rectangle is 4 units high and x + 3 units wide. Split the width into x and 3. Write its area two ways.

Lesson model



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Practice and compare methods

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P5. Simplify x + x + x + 2x. Then check your result when x = 3.

P6. Are 4(t + 2) and 4t + 2 equivalent? Check t = 1. Write a form that is equivalent to 4(t + 2).

F. Complete: 2(y + 6) = 2y + __. Check when y = 5.

D. Lena wrote 5(z + 2) = 5z + 2. Find the mistake and repair it.

X1. A packet contains 3 cards. Write the number in p packets plus 2 loose cards; evaluate at p = 2.

X2. Evaluate 2t + 3 when t = 2. Show the multiplication first.

X3. Evaluate $2^2 + 3$. Explain what the exponent means.

X4. Expand 2(x + 3) and check at x = 2.

Worked example - compare with the slide model

Find 3(2 + 4).

The parentheses name one group: 2 + 4 = 6.

Three groups give 3 × 6 = 18.

Split each group instead: 3 × 2 + 3 × 4 = 6 + 12 = 18.

Result: 18

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Challenge, explain and check

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C1. A school orders 6 kits. Each kit has x pencils and 4 markers. Write the total number of tools two ways. For x = 7, find the total. Explain why 6x + 4 is incorrect.

C2. Two rectangular banners are both 3 m high. Their widths are x m and x + 2 m. Write their combined area in a simplified form. Find it for x = 4. Can one rectangle show the same area?

Finish the lesson

S. Distributing works because each group contains __ and __.

E. Which expression equals 3(2x + 4) for every x? Choose 6x + 12, 6x + 4, or 5x + 7. Show how the 3 acts on both parts.